■ AI Student Skill Gap & Career Readiness Analyzer

Mini Project Documentation

Live Project: https://skill-mirror-path.base44.app

1. Project Overview

Many college students learn different programming languages, tools, and technologies without knowing whether those skills are actually required for their desired career. The AI Student Skill Gap & Career Readiness Analyzer helps students identify their current technical skills, skills required for a selected job role, missing or weak skills, career readiness level, recommended skills to learn, and a personalized learning direction.

2. Problem Statement

Many college students do not know which technical skills they are missing for their desired job role. They often learn random technologies without understanding the gap between their current skills and industry requirements. This project aims to provide students with a simple platform to identify their skill gaps and understand what they should learn next.

3. Proposed Solution

The application allows students to enter their educational qualification, desired job role, current technical skills, programming knowledge, and other relevant skills. The system analyzes the entered information and provides a career-readiness assessment.

```
Student Information
        ↓
Current Skills
        ↓
Desired Job Role
        ↓
Skill Comparison
        ↓
Skill Gap Analysis
        ↓
Career Readiness Score
        ↓
Recommended Skills
        ↓
Learning Direction
```

4. Key Features

Student Profile: Enter name, education, branch, year of study, and career interest.

Desired Job Role: Select or enter a target role such as Software Engineer, Python Developer, Data Analyst, AI/ML Engineer, Web Developer, or Backend Developer.

Current Skill Input: Enter technical skills currently known, such as Python, SQL, HTML, CSS, Java Basics, and DBMS.

Skill Gap Analysis: Compare current skills with skills expected for the selected career role.

Career Readiness: Provide an overall indication of career readiness based on entered skills.

Learning Recommendations: Suggest skills to focus on next, such as REST APIs, Flask, Git & GitHub, Backend Development, and Problem Solving.

Career Roadmap: Show a possible learning path from the student's current skill level toward the desired job role.

5. AI & No-Code Technology

This mini project was developed using Base44 AI, a no-code/AI-powered application development platform. AI-assisted development was used to generate and structure the web application.

Tool Used: Base44 AI

Live Project: https://skill-mirror-path.base44.app

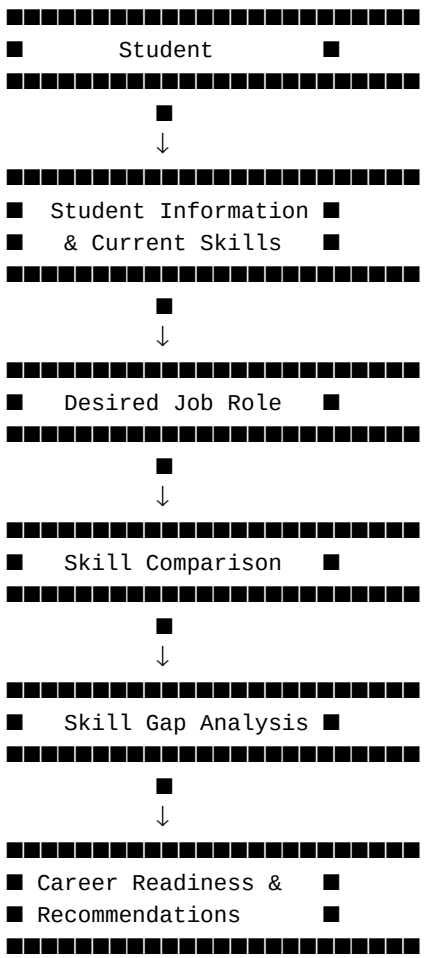
6. AI/ML Concept

The project is inspired by the concept of skill matching and gap analysis.

```
Required Skills
      -
Student's Current Skills
      =
Skill Gap
```

Example: Required skills may include Python, SQL, DBMS, Git, Flask, and APIs. If a student has only Python, SQL, and DBMS, the identified gap can include Git, Flask, and APIs.

7. System Architecture



8. Technologies / Tools Used

Technology / Tool	Purpose
Base44 AI	Application development
AI-assisted development	Feature generation and logic
Web Technologies	User interface
GitHub	Project documentation and version control
Browser	Application access and testing

9. How to Use

1. Open the project: https://skill-mirror-path.base44.app
2. Enter student information.
3. Select the desired job role.
4. Enter current technical skills.
5. Submit the information for analysis.
6. Review the skill gap, career readiness, missing skills, recommended skills, and learning direction.

10. Example Use Case

Education: B.Tech CSE-AIML
Year: 3rd Year
Target Role: Software Engineer
Current Skills: Python, SQL, DBMS, HTML, CSS, Java Basics

Possible skills to further develop include Git & GitHub, REST APIs, Backend Development, Data Structures & Algorithms, and Software Development Practices.

11. Target Users

College students, B.Tech students, Computer Science students, AI/ML students, fresh graduates, students preparing for software jobs, and students who are unsure what skills to learn.

12. Advantages

• Easy to use
• Student-focused
• Helps identify skill gaps
• Provides career direction
• Reduces random learning
• Encourages focused skill development
• Accessible through a web browser
• Demonstrates AI-assisted no-code development

13. Future Enhancements

Resume upload and analysis; advanced AI-based skill recommendations; detailed skill-wise scoring; coding assessment; technical interview preparation; HR interview practice; personalized learning resources; job recommendation based on skills; progress tracking; skill improvement milestones; LinkedIn/GitHub profile analysis.

14. Limitations

The career-readiness result should be treated as a guidance tool, not a guarantee of employment. Actual job readiness also depends on coding ability, projects, problem-solving skills, communication, interview performance, industry experience, and job-specific requirements.

15. Suggested GitHub Repository Structure

```
AI-Student-Skill-Gap-Career-Readiness-Analyzer/
■
■■■ README.md
■■■ screenshots/
■   ■■■ home-page.png
■   ■■■ skill-analysis.png
■   ■■■ career-readiness.png
■■■ project-documentation/
    ■■■ project-report.pdf
```

16. Project Information

Project Title: AI Student Skill Gap & Career Readiness Analyzer
Project Type: Mini Project
Domain: Artificial Intelligence / Career Technology
Development Approach: AI-Assisted No-Code Development
Development Tool: Base44 AI
Target Users: College Students & Fresh Graduates
Live Application: https://skill-mirror-path.base44.app

17. Developed By

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B.Tech CSE – Artificial Intelligence & Machine Learning
3rd Year

18. Conclusion

The AI Student Skill Gap & Career Readiness Analyzer provides a simple solution to a common problem faced by college students: not knowing which skills they need for their desired career. By comparing a student's current skills with the skills required for a target job role, the application helps students understand their skill gaps and focus their learning in the right direction. The project also demonstrates how AI-powered no-code development tools such as Base44 AI can be used to rapidly build useful student-focused applications.

Live Demo: https://skill-mirror-path.base44.app